# Lines and Planes in Three Dimensions

## Section 1: Vector and Cartesian Equations of Lines

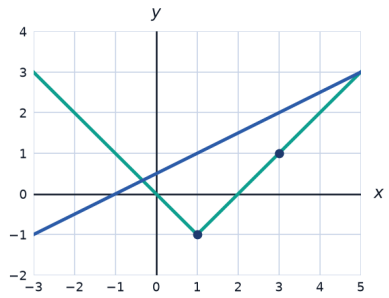
**1** Find the vector equation of the line passing through point $A(2,\ -1, 4)$ in the direction of $\mathbf{d} = [3, 5,\ -2]$.

**a.** $\mathbf{r} = [2,\ -1, 4] + \lambda[3, 5,\ -2]$

**b.** $\mathbf{r} = [3, 5,\ -2] + \lambda[2,\ -1, 4]$

**c.** $\mathbf{r} = [2 + 3\lambda,\ -1 - 5\lambda, 4 - 2\lambda]$

**d.** $\mathbf{r} = [5, 4, 2] + \lambda[1, 1, 1]$




**2** Convert the symmetric Cartesian line equations $\frac{x-3}{2} = \frac{y+1}{-4} = \frac{z-5}{1}$ into a direction vector.

**a.** $[2,\ -4, 1]$

**b.** $[3,\ -1, 5]$

**c.** $[-2, 4,\ -1]$

**d.** $[1/2,\ -1/4, 1]$

**3** Two lines in 3D that do not intersect and are not parallel are called:

**a.** Skew lines

**b.** Coplanar lines

**c.** Coincident lines

**d.** Asymptotic lines

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## Section 2: Vector and Cartesian Equations of Planes

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**4** Find the Cartesian equation of the plane passing through $P(1, 2, -3)$ with normal vector $\mathbf{n} = [2, -4, 5]$.

**a.** $2x - 4y + 5z = -21$

**b.** $2x - 4y + 5z = 21$

**c.** $x + 2y - 3z = -21$

**d.** $2x + 4y + 5z = 0$

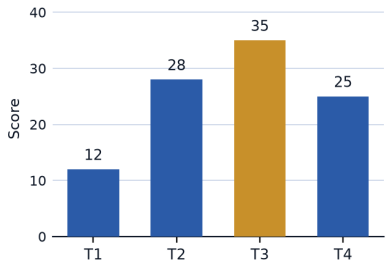
**5** What is a normal vector to the plane $3x - 7y + z = 14$?

**a.** $[3, -7, 1]$

**b.** $[3, 7, 1]$

**c.** $[14, 14, 14]$

**d.** $[-3, 7, 14]$




**6** Find a vector perpendicular to the plane containing the non-collinear vectors $\mathbf{u}$ and $\mathbf{v}$.

**a.** $\mathbf{u} \times \mathbf{v}$

**b.** $\mathbf{u} \cdot \mathbf{v}$

**c.** $\mathbf{u} + \mathbf{v}$

**d.** $\mathbf{u} - \mathbf{v}$

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## Section 3: Intersections and Angles

**7** Find the intersection point of line $\mathbf{r} = [1 + 2t, -1 + t, 3 - t]$ with plane $x + 2y - z = 4$.

**a.** $t = 1 \Rightarrow (3, 0, 2)$

**b.** $t = 0 \Rightarrow (1, -1, 3)$

**c.** $t = 2 \Rightarrow (5, 1, 1)$

**d.** No intersection

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**8** How is the acute angle $\theta$ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ calculated?

**a.** $\sin\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

**b.** $\cos\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

**c.** $\tan\theta = \frac{|\mathbf{d}\times\mathbf{n}|}{|\mathbf{d}\cdot\mathbf{n}|}$

**d.** $\theta = \arccos(\mathbf{d}\cdot\mathbf{n})$

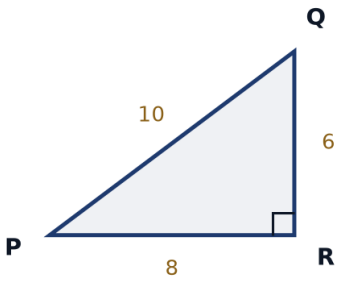
**9** Find the acute angle between two planes with normal vectors $\mathbf{n}_1 = [1, 0, 1]$ and $\mathbf{n}_2 = [0, 1, 1]$.

**a.** $\frac{\pi}{3}$ $(60^\circ)$

**b.** $\frac{\pi}{4}$ $(45^\circ)$

**c.** $\frac{\pi}{2}$ $(90^\circ)$

**d.** $\frac{\pi}{6}$ $(30^\circ)$




## Section 4: Shortest Distances in 3D

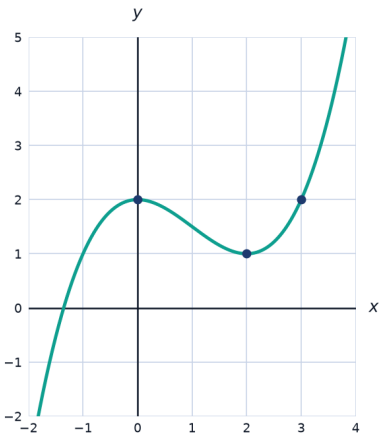
**10** What is the shortest distance from point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz = D$?

**a.** $\frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$

**b.** $\frac{|Ax_0 + By_0 + Cz_0 + D|}{A^2 + B^2 + C^2}$

**c.** $\sqrt{(x_0 - A)^2 + (y_0 - B)^2 + (z_0 - C)^2}$

**d.** $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$




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**11** Find the distance from the origin $(0, 0, 0)$ to the plane $2x - 3y + 6z = 28$.

**a.** $4$

**b.** $28$

**c.** $\frac{28}{7} = 4$

**d.** $\frac{28}{49}$

**12** How is the shortest distance between two parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 14$ calculated?

**a.** $\frac{|14 - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{9}{3} = 3$

**b.** $\frac{14 + 5}{3} = \frac{19}{3}$

**c.** $14 - 5 = 9$

**d.** $\frac{9}{9} = 1$

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